

Malik Tafheem Ul Islam

maliktafheem10@gmail.com [linkedin.com/in/maliktafheem](https://www.linkedin.com/in/maliktafheem) github.com/maliktafheem maliktafheem.github.io

Research interests

Evaluation and reliability of generative and multimodal models, especially **rule and instruction following in video generation and editing**, **multimodal reasoning**, and **VLM-as-judge evaluation**.

Education

National Institute of Technology Srinagar

Aug 2025 to Jun 2027

M.Tech in Computer Science and Engineering, **CGPA 9.18/10** (expected)

- Courses: Machine Learning (A+), Advanced Algorithms (A+), Introduction to Data Science (A+), Artificial Intelligence and Fuzzy Logic (A+), Computer Vision (A), Network & System Security (A), Real Time Operating Systems (A)
- In progress (Autumn 2026): Deep Learning, Advanced Theory of Computation, Big Data, Research Methodology

National University of Sciences and Technology (NUST), Islamabad

Sep 2019 to Jun 2023

BS in Computer Science, **CGPA 3.64/4.00**

- Courses: Linear Algebra, Probability and Statistics, Numerical Analysis, Data Structures and Algorithms, Design and Analysis of Algorithms, Artificial Intelligence, Digital Image Processing, Advanced Topics in Computing (computer vision), Theory of Automata, Operating Systems, Computer Networks

Research and professional experience

Research Assistant, video generation models

Dec 2025 to present

GAASH Lab, NIT Srinagar / Srinagar, India

- Study **reasoning and instruction following** in video generation models, including temporal, spatial, and causal consistency.
- Build **reproducible inference and validation pipelines** for video generation and editing models.
- Build **evaluation workflows** that combine automatic scoring with human review to check consistency and instruction following in generated videos.

Lab Assistant

Sep 2025 to present

NIT Srinagar / Srinagar, India

- Assist in the Python Programming, Artificial Intelligence, and Network & Security labs: demonstrate concepts (OOP and NumPy/pandas; search, minimax, and CSPs; TCP/IP and packet analysis), help students debug their programs, and assess their work.

Research Intern

Dec 2025 to Feb 2026

IIT Roorkee, Dept. of Electrical Engineering / 6 weeks, supervised by Dr. Abdul Saleem Mir

- Replicated the **T2I-CompBench++** benchmark end to end, covering **all 8 sub-categories** of compositional text-to-image evaluation (attribute binding, spatial and non-spatial relations, numeracy, and complex compositions).
- Ran **chain-of-thought and self-consistency** prompting on open LLMs and VLMs in Python and compared the outputs qualitatively with direct prompting.
- Surveyed work on reasoning-enhanced generation in language, vision-language, and diffusion models.

Machine Learning Engineer

Jul 2024 to Jan 2025

Freelance

- Built **image classifiers** with CNNs and linear models, and clustering workflows with PCA and K-means.
- Wrote **TensorFlow and PyTorch** pipelines for preprocessing, model tuning, visualization, and interpretation.

Research Assistant

Apr 2023 to Jun 2024

System Analysis and Verification (SAVe) Lab, NUST

- Studied **image segmentation for collision avoidance in surgical robots** and evaluated segmentation methods on surgical datasets.
- Implemented **NSGA-II**-based image segmentation in Java.
- Explored U-Net, DeepLab, and Mask R-CNN for segmentation in robotics.

Software Engineering Intern

May 2023 to Nov 2023

Bitnine Global Inc. (remote)

- Developed **C/C++ utilities** for LDAP synchronization and data loading in AgensSQL.
- Configured ldap2pg synchronization and analyzed pgloader data structures for the loader.

Research Assistant

Jun 2021 to May 2022

TUKL-NUST R&D Center

- Reviewed detection and tracking methods, including ByteTrack, FairMOT, YOLO, and EfficientDet.
- Built an **annotated fish dataset** with Supervisely, Roboflow, RectLabel, and LabelImg.
- Applied **transfer learning to YOLO detectors** for fish detection in a tracking pipeline.

- Compared logistic regression and SVM for ride matching and reached **82% accuracy** after feature engineering.

Publications and manuscripts

Intent-Based Networking and Agentic AI for Next-Generation Network Control: A Systematic Survey, Taxonomy, and Research Agenda

Malik Tafheem Ul Islam, Iram Naseer, Mir Salim Ul Islam

Telecommunication Systems (Springer). **Under review** (major revision resubmitted, Sep 2026).

A benchmark for rule-based instruction following in video generation and editing

Malik Tafheem Ul Islam*, Iram Naseer*, Tajamul Ashraf* (*equal contribution)

In preparation.

Projects

KATHE 2026 shared task: English-to-Kashmiri machine translation

[GitHub](#) 

- Team of two, with Iram Naseer. **Ranked 7th of 38 teams** on the private test set with a fine-tuned **IndicTrans2 (1.1B)** model for English to Kashmiri in the Perso-Arabic script. Public leaderboard score 13.90 ($\sqrt{\text{BLEU} \times \text{chrF++}}$).
- Fixed a **preprocessing error** in IndicTrans2 that deleted Arabic short-vowel marks from Kashmiri text. Released the weights on Hugging Face.

Anti-drone system using low-cost sensors and machine learning

Final-year project, NUST

- Trained YOLOv5/v7/v8 and **YOLO-NAS** detectors on visible-light and infrared drone data, reaching **about 90% mAP**.
- Added **SORT** and **DeepSORT** for real-time multi-object tracking, and field-tested detection with a smartphone camera in controlled and outdoor conditions.

House price prediction

[GitHub](#) 

- Explored the data and engineered features; an MLP regressor reached **$R^2 = 0.921$** on a 30% test split. Built a Streamlit interface for predictions.

Image colorization with conditional GANs

[GitHub](#) 

- Built a **conditional GAN** with a U-Net/ResNet-18 generator that predicts color channels from grayscale images, trained on about 10,000 COCO images with L1 and adversarial losses in Lab color space.

Web of Knowledge

[GitHub](#) 

- Built a **C++ bibliographic database** with AVL trees, linked lists, and graphs, with publication and co-author queries, collaboration checks, and graph-distance search.

Skills

- **Research areas:** generative AI, video generation and editing, multimodal reasoning, computer vision, model evaluation, agentic AI for network control
- **Programming:** Python, C/C++, Java, SQL, R, MATLAB, Assembly
- **ML and data:** PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, pandas; CNNs, GANs, YOLO, U-Net, DeepLab, Mask R-CNN
- **Compute and models:** HPC with Slurm (H100 MIG), lab GPUs (A10, L4); 12+ open video generation models; VLM-as-judge pipelines (Qwen3-VL); Hugging Face
- **Tools:** Git, Docker, LaTeX, Streamlit, Supervisely, Roboflow, RectLabel, LabelImg
- **Systems:** Linux (Ubuntu, Kali), Windows

Honors

- GATE 2025, Data Science and Artificial Intelligence: **All India Rank 1388**
- SAT Subject Tests: Mathematics **800/800**, Physics 790/800, Chemistry 790/800
- **Merit-based scholarship** at NUST, awarded on GPA

Certifications

Deep Learning Specialization  (DeepLearning.AI); Machine Learning  (Stanford Online); Associate AI Engineer for Data Scientists  (DataCamp, 40-hour track)